COMP 372/471 Lecture Notes

Release 1.0

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COMP 372 / COMP 471
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Welcome! This version of the course is normally taught by Konstantin Läufer and focuses on functional and concurrent programming languages with examples and projects mostly in Scala.

We maintain these lecture notes using GitHub. We also offer PDF and ePub formats for offline/mobile reading.

Warning: These notes are still being written, so expect a few rough edges. But we're getting closer!

CHAPTER

ONE

CONTENTS

1.1 Overview

In this chapter, we provide an outline of the course, along with a brief motivation and key learning outcomes.

1.1.1 Motivation

There are a number of reasons why you would want to study programming languages. To name a few:

- Knowing several languages will help you solve more problems in better ways.
- The subject matter is exciting and intellectually challenging.
- Google, Meetup, and increasing numbers of local startups love it (from an employer's point of view)!

1.1.2 Key learning outcomes

We deliberately chose the learning outcomes for this course to complement other courses in your computer science or software engineering curriculum. Upon entering this course, you will already be familiar with the imperative and object-oriented paradigms, coming from languages such as Java, C++, C#, or Python. Most likely, you will also be familiar with basic machine architecture and memory management using C or C++.

In this course, we will pursue one overarching learning outcome and three more specific ones:

- an understanding of the programming language design space
- proficiency in functional programming
- an understanding of program representation and interpretation/execution
- basic competence in concurrent programming

Given the widespread availability in multi-core hardware ranging from mobile devices to enterprise servers, there is a growing interest in programming paradigms and techniques that will take advantage of this hardware. Functional programming, which de-emphasizes mutable state, in conjunction with suitable concurrency constructs, is particularly useful. We will also "look under the hood" by studying how to represent programs and interpret, execute, or otherwise process those representations.

1.1.3 Some starting points for functional programming

You might find the following resources useful to get into the functional programming mindset:

- Presentation by Martin Odersky: Scala with Style (YouTube)

- Interview with Simon Peyton-Jones on Functional Programming and Haskell (SE Radio audio podcast)
- Presentation by Läufer, Thiruvathukal, and Kaylor: The Promise of Statically Typed Functional Programming: Broader Context and Language Comparison (visual presentation)

Todo

incorporate visual presentation here

1.1.4 Course outline

Here is a more detailed course outline (subject to revision).

- functional programming (6 weeks)
  - defining algebraic data types
    - scalars: enumerations
    - sublinear structures: numbers, option
    - linear structures: lists, maps
    - nonlinear structures: trees
  - implementing behaviors on algebraic data types
    - pattern matching
    - recursion
  - higher-order functions
  - predefined types and their behaviors
  - recursion patterns
  - higher-kinded types
- programming language representation and interpretation/execution (5 weeks)
  - language design space
  - scanners (lexical analyzers)
  - parsers (syntax analyzers)
  - interpreters and compilers
  - domain-specific languages
- concurrent and parallel programming (4 weeks)
  - parallel collections
  - futures and promises
  - progress reporting and cancelation
  - asynchronous programming/reactive extensions (Rx)
  - advanced mechanisms
    - explicit threads
    - actors

    * software-transactional memory
    * task-parallel library
    * functional data structures

1.1.5 Bloom's taxonomy

The letters refer to the successive levels of learning from the cognitive domain of Bloom's taxonomy:

- K: know/remember the term
- C: comprehend/understand the concept
- A: apply the technique
- analyze/evaluate/create (advanced levels)

Todo

map Bloom levels to course outline

1.2 Background

- requirements
- key principles, including orthogonality/separation of concerns
- case study: processtree

1.3 The Functional Programming Paradigm

In this chapter, we study the functional programming paradigm, with examples and projects mostly in Scala.

The orthogo

processtree example in detail

refer to "detailed outline"

- project 0b and 1 clarifications - [echotest-scala](https://github.com/lucproglangcourse/echotest-scala) example - TDD styles in Scala
- project 0c and 0d reminders
- [overview talk on PL history and paradigms](http://klaeufer.github.com/luc-amc.html) - Scala version of Haskell example from talk - [shootout benchmark](http://benchmarksgame.alioth.debian.org) - process tree [assignment](http://osdi.etl.luc.edu/homework/home-lab-assignment-1) and [examples in various languages](https://github.com/lucproglangcourse?q=processtree) including [Scala](https://github.com/lucproglangcourse/processtree-scala) - also guiding example for project 1
- console applications - role as composable building blocks in Unix (using pipes) - relationship with functional programming - importance of constant-space complexity
- scripting-style Scala: solving problems using built-in types and type constructors - tuples - collections

  - [*Option[_]*](scala-lang.org/api/current/index.html#scala.Option) and [*Try[_]*](http://scala-lang.org/api/current/index.html#scala.util.Try)
  - [*List[_]*](http://scala-lang.org/api/current/index.html#scala.collection.immutable.List) and [*Map[_, _]*](http://scala-lang.org/api/current/index.html#scala.collection.immutable.Map)
  - key methods: *map*, *filter*/*withFilter*, *find*, *flatMap*, *sum*, *fold*, *groupBy*
  - *for* comprehensions
  - [misc-explorations-scala](https://github.com/lucproglangcourse/misc-explorations-scala) examples
- object-oriented Scala (see also Swarts chapters 8 and 9) - can start by using Scala as a "better Java" - various improvements, including
  - [unified types](http://docs.scala-lang.org/tutorials/tour/unified-types.html)
  - [standalone objects](http://docs.scala-lang.org/tutorials/tour/singleton-objects.html)
  - [case classes](http://docs.scala-lang.org/tutorials/tour/case-classes.html) and [pattern matching](http://docs.scala-lang.org/tutorials/tour/pattern-matching.html)
  - [traits](http://docs.scala-lang.org/tutorials/tour/traits.html) (generalization of interfaces and restricted form of abstract classes, can be combined)
  - package structure decoupled from folder hierarchy
  - we'll study these features as we encounter them
  - examples below after discussing the next topic
- algebraic data types - fundamental building blocks: (disjoint) sum, product, arity, recursion, type parameters

  Shape = Circle(Int)

  Rectangle(Int, Int)
  Group(Seq(Shape))
  Location(Int, Int, Shape)

  - nonrecursive, e.g. finite enumerations, boolean, option, try
  - sublinear, e.g. natural numbers
  - linear, e.g. list, map
  - nonlinear, e.g. tree, many custom domain models
  - separation of concerns - structure - content - traversal - processing
  - parametric polymorphism (genericitiy) as an orthogonal mechanism
  - expressing algebraic data types in an object-oriented language
  - key predefined collection data types *are* algebraic data types - [*Option[_]*](scala-lang.org/api/current/index.html#scala.Option) and [*Try[_]*](http://scala-lang.org/api/current/index.html#scala.util.Try) - [*List[_]*](http://scala-lang.org/api/current/index.html#scala.collection.immutable.List) and [*Map[_, _]*](http://scala-lang.org/api/current/index.html#scala.collection.immutable.Map)
  - examples - process tree [original assignment](http://osdi.etl.luc.edu/homework/home-lab-assignment-1) and [examples](https://github.com/lucproglangcourse?utf8=%E2%9C%93&query=processtree) - [misc-explorations-scala](https://github.com/lucproglangcourse/misc-explorations-scala)
- project 1 discussion: - imperative/mutable versus pure functional/immutable - modularization - [processtree-scala](https://github.com/lucproglangcourse/processtree-scala) guiding example
  - algebraic data types (continued)

  - key predefined collection data types - [*Option[_]*](scala-lang.org/api/current/index.html#scala.Option) and [*Try[_]*](http://scala-lang.org/api/current/index.html#scala.util.Try) - [*List[_]*](http://scala-lang.org/api/current/index.html#scala.collection.immutable.List) and [*Map[_, _]*](http://scala-lang.org/api/current/index.html#scala.collection.immutable.Map) - key methods: *map*, *filter*/*withFilter*, *find*, *flatMap*, *sum*, *fold*, *groupBy* - *for* comprehensions
  - pattern matching in practice
  - examples - [expressions-oo-java](https://github.com/LoyolaChicagoCode/misc-java/blob/master/src/main/java/expressions/SimpleExpressions.java) - [expressions-visitor-java](https://github.com/LoyolaChicagoCode/misc-java/blob/master/src/main/java/vexpressions/VisitorExpressions.java) - [expressions-scala](https://github.com/lucproglangcourse/expressions-scala) - [shapes-android-java](https://github.com/LoyolaChicagoCode/shapes-android-java) - [shapes-oo-scala project skeleton](https://github.com/lucproglangcourse/shapes-oo-scala) - process tree [original assignment](http://osdi.etl.luc.edu/homework/home-lab-assignment-1) and [examples](https://github.com/lucproglangcourse?utf8=%E2%9C%93&query=processtree) - [misc-explorations-scala](https://github.com/lucproglangcourse/misc-explorations-scala)
- project 2a discussion
- TDD styles in Scala and [echotest example](https://github.com/lucproglangcourse/echotest-scala)
- (time permitting) brief heads-up on continuous integration/continuous delivery - https://travis-ci.org/LoyolaChicagoCode/processtree-scala - https://coveralls.io/github/LoyolaChicagoCode/processtree-scala - https://github.com/LoyolaChicagoCode/scalamu - https://www.codacy.com/public/laufer/scalamu/dashboard - http://issuestats.com/github/LoyolaChicagoCode/scalamu
  - algebraic data types (continued)
  - algebraic data types in practice - case classes/objects - pattern matching - built-in methods - external behaviors ([Visitor pattern](https://sourcemaking.com/design_patterns/visitor))
  - examples - [expressions-oo-java](https://github.com/LoyolaChicagoCode/misc-java/blob/master/src/main/java/expressions/SimpleExpressions.java) - [expressions-visitor-java](https://github.com/LoyolaChicagoCode/misc-java/blob/master/src/main/java/vexpressions/VisitorExpressions.java) - [expressions-scala](https://github.com/lucproglangcourse/expressions-scala) - [shapes-android-java](https://github.com/LoyolaChicagoCode/shapes-android-java) - [shapes-oo-scala project skeleton](https://github.com/lucproglangcourse/shapes-oo-scala) - process tree [original assignment](http://osdi.etl.luc.edu/homework/home-lab-assignment-1) and [examples](https://github.com/lucproglangcourse?utf8=%E2%9C%93&query=processtree) - [misc-explorations-scala](https://github.com/lucproglangcourse/misc-explorations-scala) org chart examples
  - tuple versus sequence/list
  - *map* versus *flatMap*
  - *for* comprehensions
  - console applications as composable building blocks in Unix (using pipes)
  - importance of constant-space complexity for scalability
- recap of predefined list operations - keep in mind that [lists are head/tail-optimized](http://www.scala-lang.org/api/current/index.html#scala.collection.immutable.List) - observations
  - *foldLeft* is usually what we want: *linear-time* and *constant-space* (tail-recursive)
  - *foldRight* is *linear-time* and *linear-space* (*not* tail-recursive) but goes with the natural head-tail structure of the list
  - *xs.foldRight(z)(f) == xs.reverse.foldLeft(z)(g)* where g is f with the arguments switched

  - Scala library implementations of these functions - find desired method in documentation, expand, and look at *definition classes* - these tend to appear more complex than expected for performance reasons - [*foreach*](https://github.com/scala/scala/blob/v2.11.7/src/library/scala/collection/immutable/List.scala#L378) - [*foldLeft*](https://github.com/scala/scala/blob/v2.11.7/src/library/scala/collection/TraversableOnce.scala#L153) - [*reverse*](https://github.com/scala/scala/blob/v2.11.7/src/library/scala/collection/immutable/List.scala#L386) - [*foldRight*](https://github.com/scala/scala/blob/v2.11.7/src/library/scala/collection/immutable/List.scala#L396) - [*map*](https://github.com/scala/scala/blob/v2.11.7/src/library/scala/collection/immutable/List.scala#L270) - [*length*](https://github.com/scala/scala/blob/v2.11.7/src/library/scala/collection/LinearSeqOptimized.scala#L49)
  - *challenge*: can we write (efficiently or not) - *length*, *sum*, *reverse*, *filter*, *find*, *map* as fold? - fold as *map*?!?
  - efficiency considerations in more detail -> next week - [space complexity, tail recursion, and accumulators](http://blog.emielhollander.nl/tail-recursion-and-the-accumulator-in-scala/) - [tail recursion, trampolines, and continuations](http://blog.richdougherty.com/2009/04/tail-calls-tailrec-and-trampolines.html) (advanced)
  - observations - *fold* handles the *traversal* concern and separates it from the *processing* concern (goes a step further than the visitor pattern) - domain and codomain of *fold*? - domain and codomain of *map*?
- writing our own (recursive) functions on options, lists, streams, and other algebraic data types -> next week - examples
  - [factorial](https://github.com/LoyolaChicagoCode/misc-explorations-scala/blob/master/factorial.sc)
  - [functions on lists](https://github.com/LoyolaChicagoCode/misc-explorations-scala/blob/master/functionsOnLists.sc)
  - [functions on streams](https://github.com/lucproglangcourse/misc-explorations-scala/blob/master/functionsOnStreams.sc)
  - [functions on simple natural numbers](https://github.com/lucproglangcourse/misc-explorations-scala/blob/master/nat.sc)
  - [expressions-scala](https://github.com/lucproglangcourse/expressions-scala)
  - observations - the Y-combinator handles the *recursion* concern *for behaviors* and separates it from the concern of what should happen in each step of the recursion
- formalizing algebraic data types as initial F-algebras - concepts -> next week
  - [(endo)functor](https://hseeberger.wordpress.com/2010/11/25/introduction-to-category-theory-in-scala)
  - [F-algebra](https://www.fpcomplete.com/user/bartosz/understanding-algebras)
  - fold = catamorphism
  - the μ-combinator handles the *recursion* concern *for structures* and separates it from the nature of the structure itself
  - F-coalgebra
  - unfold = anamorphism
  - practical applications -> next week
  - examples - [expressions-scala](https://github.com/lucproglangcourse/expressions-scala) versus [expressions-algebraic-scala](https://github.com/lucproglangcourse/expressions-algebraic-scala) - [Scalaμ library](https://github.com/lucproglangcourse/scalamu) and [examples](https://github.com/lucproglangcourse/scalamu/examples)
- project 2b discussion

(time permitting)

- TDD styles in Scala and [echotest example](https://github.com/lucproglangcourse/echotest-scala)
- brief heads-up on continuous integration/continuous delivery -> probably next week - https://travis-ci.org/LoyolaChicagoCode/processtree-scala - https://coveralls.io/github/LoyolaChicagoCode/processtree-scala - https://github.com/LoyolaChicagoCode/scalamu - https://www.codacy.com/public/laufer/scalamu/dashboard - http://issuestats.com/github/LoyolaChicagoCode/scalamu

References

- [Understanding F-Algebras](https://www.fpcomplete.com/user/bartosz/understanding-algebras)
- [Gibbons](https://www.cs.ox.ac.uk/jeremy.gibbons/publications/origami.pdf) (advanced)
- [Oliveira & Cook](http://www.cs.utexas.edu/~wcook/Drafts/2012/ecoop2012.pdf) (advanced)
- efficiency considerations for predefined list operations in more detail - [space complexity, tail recursion, and accumulators](http://blog.emielhollander.nl/tail-recursion-and-the-accumulator-in-scala/) - [tail recursion, trampolines, and continuations](http://blog.richdougherty.com/2009/04/tail-calls-tailrec-and-trampolines.html) (advanced)
- writing our own (recursive) functions on options, lists, streams, and other algebraic data types - examples
  - [factorial](https://github.com/LoyolaChicagoCode/misc-explorations-scala/blob/master/factorial.sc)
  - [functions on lists](https://github.com/LoyolaChicagoCode/misc-explorations-scala/blob/master/functionsOnLists.sc)
  - [functions on streams](https://github.com/lucproglangcourse/misc-explorations-scala/blob/master/functionsOnStreams.sc)
  - [functions on simple natural numbers](https://github.com/lucproglangcourse/misc-explorations-scala/blob/master/nat.sc)
  - [expressions-scala](https://github.com/lucproglangcourse/expressions-scala)
  - observations - the Y-combinator handles the *recursion* concern *for behaviors* and separates it from the concern of what should happen in each step of the recursion
- formalizing algebraic data types as initial F-algebras - concepts
  - [(endo)functor](https://hseeberger.wordpress.com/2010/11/25/introduction-to-category-theory-in-scala): type constructor that satisfies *identity* and *composition* laws
  - the μ-combinator handles the *recursion* concern *for structures* and separates it from the nature of the structure itself
  - fold = catamorphism *cata* for breaking down a data structure to a result value
  - [F-algebra](https://www.fpcomplete.com/user/bartosz/understanding-algebras): argument to fold, has functor F and carrier object (= result type of the fold)
  - *unfold* = anamorphism for building up a data structure
  - F-coalgebra: argument to unfold (generator), also has functor F and carrier object (= type of seed and generated values wrapped in functor)
  - practical applications
  - examples - [expressions-scala](https://github.com/lucproglangcourse/expressions-scala) versus [expressions-algebraic-scala](https://github.com/lucproglangcourse/expressions-algebraic-scala) - [Scalaμ library](https://github.com/lucproglangcourse/scalamu) and [examples](https://github.com/lucproglangcourse/scalamu/examples)
- [project 2b](https://trello.com/c/EihkYkGP/59-project-2b) clinic
- project 3a overview

- 16:30-16:45 student presentation on C++
- 16:45-17:15 presentation by alumna and part-time faculty member María Sáenz

(time permitting)

- TDD styles in Scala and [echotest example](https://github.com/lucproglangcourse/echotest-scala)
- brief heads-up on continuous integration/continuous delivery -> probably next week - https://travis-ci.org/LoyolaChicagoCode/processtree-scala - https://coveralls.io/github/LoyolaChicagoCode/processtree-scala - https://github.com/LoyolaChicagoCode/scalamu - https://www.codacy.com/public/laufer/scalamu/dashboard - http://issuestats.com/github/LoyolaChicagoCode/scalamu
- recap: formalizing algebraic data types as initial F-algebras - concepts
  - [(endo)functor](https://hseeberger.wordpress.com/2010/11/25/introduction-to-category-theory-in-scala): type constructor that satisfies *identity* and *composition* laws
  - the μ-combinator handles the *recursion* concern *for structures* and separates it from the nature of the structure itself
  - fold = catamorphism *cata* for breaking down a data structure to a result value
  - [F-algebra](https://www.fpcomplete.com/user/bartosz/understanding-algebras): argument to fold, has functor F and carrier object (= result type of the fold)
  - *unfold* = anamorphism for building up a data structure
  - F-coalgebra: argument to unfold (generator), also has functor F and carrier object (= type of seed and generated values wrapped in functor)
  - practical applications
  - examples - [expressions-scala](https://github.com/lucproglangcourse/expressions-scala) versus [expressions-algebraic-scala](https://github.com/lucproglangcourse/expressions-algebraic-scala) - [project 2a](https://trello.com/c/nPaqQAJf/58-project-2a) versus [project 2b](https://trello.com/c/EihkYkGP/59-project-2b)
  - key insights - commonalities among recursive types
    * μ-based: *Nat*, *Expr*, *Shape*, etc.
    * *Cofree*-based: *List*, *Tree*, *OrgChart*, etc.
    * how are, say, *Option*, *List*, and *Tree* related?
    * structural
    * behavioral
    * potential for code reuse - [Scalaμ library](https://github.com/lucproglangcourse/scalamu) - [Scalaz library](https://github.com/scalaz/scalaz)

1.4 Program Representation and Interpretation

- programming language [compilation](http://en.wikipedia.org/wiki/Compiler)/{[}interpretation{]}(http://en.wikipedia.org/wiki/Interpreter_(computing)) toolchain phases overview - source code (string stored in file) - *lexical analysis* - sequence of tokens - *syntax analysis (parsing)* - abstract syntax tree - *static checks* - *optimization* - transformed abstract syntax tree - *code generation* - machine code or byte code - *execution* - alternative: *direct interpretation of abstract syntax trees*
- [project 3a](https://trello.com/c/1Eq8r4vs/27-project-3a) discussion

Presentation

According to this https://trello.com/c/a7vQg3qp/54-presentation-schedule.

References

- [Mogensen](http://www.diku.dk/hjemmesider/ansatte/torbenm/Basics)
- compiler front-end in more detail - lexical analysis: regular expressions, NFA, DFA (Mogensen ch. 2) - syntax analysis: context-free grammars, EBNF, parsing (Mogensen ch. 3)
- programming language semantics - [static semantics](http://en.wikipedia.org/wiki/Programming_language#Semantics) - compile-time
  - flow analysis
  - typing
  - [dynamic semantics](http://en.wikipedia.org/wiki/Semantics_of_programming_languages) - run-time - denotational - operational, e.g. our interpreters - axiomatic
  - simple imperative language example - [operational semantics](http://plone.cs.luc.edu/laufer-archived/teaching/473/handouts/SimpleImperative.html) - [Java implementation](https://github.com/LoyolaChicagoCode/misc-java/blob/master/src/main/java/imperative/SimpleImperative.java) - [Scala implementation](https://github.com/LoyolaChicagoCode/misc-scala/blob/master/src/main/scala/imperative/)
- interpreter pattern - http://c2.com/cgi/wiki?InterpreterPattern - http://en.wikipedia.org/wiki/Interpreter_pattern

1.4.1 A simple imperative language

In this section, we discuss the design of an interpreter and programming environment for a simple imperative language.

The accompanying lecture is available as a set of screencasts:

- [foundations](https://youtu.be/-bDcsvlY5oA)
- [implementation](https://youtu.be/mj-tq_kdeFU)

Objectives

We are leading toward building an interpreter for a language resembling JavaScript with object-like features. We are doing this with two aims in mind:

- To understand an object-oriented programming language from the "inside".
- To understand better the object-oriented and functional design patterns that we have learned by using them in the context of a small but somewhat sophisticated program.

Syntax

Our language has the following features:

Integer expressions given by the BNF grammar:

e ::= variable

const
e1 + e2
e1 - e2

Statements given by the BNF grammar:

S ::= x = e

S1 ; S2
while (e) S

In the context of the lectures so far, we have made the following changes. We have added variables to expressions, thus we can handle cases like "x + 3", whereas earlier we could only write expressions such as "4 + 3". We have also introduced the assignment statement as a way to change the contents of a variable. In addition, we allow statements to be put in sequence. We also permit simple while expressions, where the guard is an expression and the loop body is executed while the gurad expression evaluates to a non-zero integer value.

Structured operational semantics (SOS)

We first formalize the intuitive execution semantics of the toy language. The point of doing this is to present the basic ideas in the interpreter without getting tied up in the programming details of the interpreter. In any case, these details are presented later in this lecture. Since our toy language has variables, we need to keep track of the state of variables. We view variables as objects with two capabilities:

- *v.get()* returns the current value of the variable
- *v.set(int i)* changes the current value of the variable to that of *i*

We think of the state of the program (memory store), which we write *M*, as a map that associates identifiers with variable objects.

The rules for evaluating expressions are quite simple.

- Evaluating constant *c*. Every constant evaluates to itself.
- Evaluating a variable whose name is *x*: Retrieve the variable object (say *v*) associated with *x* from the memory store *M*, by using *M(x)*. The required result is computed by invoking *v.get()*.
- Evaluating *e1 + e2*: Evaluate *e1* first, say to yield value *r1*. Evaluate *e2* next, say to yield value *r2*. The required result is the numeric value *r1 + r2*.
- Evaluating *e1 - e2*: Evaluate *e1* first, say to yield value *r1*. Evaluate *e2* next, say to yield value *r2*. The required result is the numeric value *r1 - r2*.

The evaluation rules are written out precisely in the following picture.

The rules for executing statements is as follows. In contrast to expression evaluation, statement execution does not yield a result. The primary consequence of executing a statement is the side effect on the store, ie. changes in the values of variables.

- Executing an assignment statement: Consider the assignment statement *x = e*. Here *e* is an expression. The steps are as follows: First, evaluate the expression *e* to yield a result, say *r*. Next, retrieve the variable object (say *v*) associated with *x* from the memory store *M*, by using *M(x)*. Perform *v.set(r)* to change the value of the variable object.
- Executing a sequence of statements *S1; S2*: Execute *S1* first. When that terminates, execute *S2*.
- Executing *while (e) do S*: Evaluate the condition *e* to yield a result *r*. If *r* is zero, the execution terminates. Otherwise, execute *S* and repeat the process.

The execution rules are given in the following pictures:

Note that the connection between the various statements is that they share a single store, ie. in the sequence of statements "x = 2; y = x + 1", the second reference to x reflects the effect of the first assignment because of the (shared) store between the two assignment statements.

The interpreter program

We now go ahead and write the interpreter program. The entire code is available as part of the [misc-scala](https://github.com/lucproglangcourse/misc-scala/tree/master/src/main/scala/imperative) example. A similar example but based on F-algebras and with better runtime error handling is also available: [simpleimperative-algebraic-scala](https://github.com/LoyolaChicagoCode/simpleimperative-algebraic-scala).

1.4.2 A simple imperative language with records

In this handout, we discuss the design of an interpreter and programming environment for a simple imperative language with records (like classes with public fields and without methods).

Objectives

We are leading toward building an interpreter for a language resembling JavaScript with object-like features. We are doing this with two aims in mind:

- To understand an object-oriented programming language from the "inside".
- To understand better the object-oriented and functional design patterns that we have learned by using them in the context of a small but somewhat sophisticated program.

We now consider the changes that arise from the addition of records. Thus, we permit:

- declaration of record types
- creation of new records of a given record type
- selection of record fields
- use of records on the left and right hand side of expressions

Syntax

Our language has the following features:

The syntactic feautures of our language are captured by the following grammar. For motivation, the sort of program that we are interested is exemplified by:

> **StudentCourseRecord = record** int firstExamScore; int secondExamScore; int totalScore;
>
> end;
>
> **StudentSemRecord = record** StudentCourseRecord course1; StudentCourseRecord course2;
>
> end;
>
> StudentSemRecord r = new StudentSemRecord(); r.course1 = new StudentCourseRecord(); r.course1.firstExamScore = 25; r.course1.secondExamScore = 35; r.course1.totalScore = r.course1.firstExamScore + r.course1.secondExamScore;
>
> r.course2 = r.course1;

In the C language, such things are known as structs. In familiar object-oriented terminology, we can think about them in this way:

- record types are classes whose only members are public member variables
- records are objects
- fields are public member variables

The record type definitions in the previous example would look as follows in Java, and the rest of program would look the same.

```
class StudentCourseRecord { public int firstExamScore; public int secondExamScore; public int totalScore;
}
class StudentSemRecord { public StudentCourseRecord course1; public StudentCourseRecord course2;
}
```

Formally, we proceed via the following BNF grammars. To simplify life for us, we will ignore type information. In this BNF grammar, we are a little bit more careful to separate L(eft) values and R(ight) values. L-values are those that can appear on the left hand side of an assignment statement, and R-values are those that appear on the right hand side of an assignment.

Record definitions are given by the BNF grammar:

```
Defn ::= record
            FieldList
        end
FieldList ::= fieldName, FieldList
        fieldName
```

L-values (fields selected from records, as well as variables) are given by the BNF grammar:

```
Lval ::= e.fieldName
        variable
```

Expressions (R-values) are given by the BNF grammar:

```
e ::= new C
        Lval
        const
        e1 + e2
        e1 - e2
```

Statements given by the BNF grammar:

```
S ::= Lval = e
        S1; S2
        while (e) do S
```

We first formalize the intutive execution semantics of the toy language. As before, the point of doing this is to present the basic ideas in the interpreter without getting tied up in the programming details of the interpreter. In any case, these details are presented later in this lecture. In particular, in this initial first cut, we will begin by ignoring declarations. Also, in this new presentation

Recall that we viewed variables as objects with two capabilities:

- get() returns the current value of the variable
- set(int x) changes the current value of the variable to that of x

Records are thought of in a similar light.

As before, we think of the state of the program, which we write S, as a map that associates identifiers with variable objects. Furthermore, as before, we distinguish evaluation and execution. In evaluation, there are two subcases, evaluating to an L-value and evaluating to an R-value.

There are two ways of having L-values. One is via variables and the second is via field selection.

- The L-value associated with a variable name is the associated variable object.
- The L-value associated with a selection e.f is obtained by first evaluating the expression e to an R-value, say r. Next, lookup on the record r with field name f is used to get the desired L-value.

These evaluation rules are written out precisely in the following picture.

Our earlier rules for evaluating R-values are presented again below..

- Evaluating an L-value. In our setup, every L-value (say l) is a variable object that is obtained from the store M. Execute l.get() to compute the return value. This rule subsumes the earlier case for variables.
- Evaluating e1 + e2: Evaluate e1 first, say to yield value v1. Evaluate e2 next, say to yield value v2. The required result is v1 + v2.
- Evaluating e1 - e2: Evaluate e1 first, say to yield value v1. Evaluate e2 next, say to yield value v2. The required result is v1 - v2.
- Evaluating constant c. Every constant evaluates to itself.

The rules for executing statements are as follows. They are similar to the ones seen before. The primary consequence of executing a statement still is the side effect on the store, ie. changes in the values of variables.

- Executing an assignment statement L = e. Here L is an L-valued expression and e is an R-valued expression. The steps are as follows. First, evaluate the expression L to yield an L-value, say l. Next, evaluate the expression e to yield an R-value, say v. Next, use l.set(v) to change the value of the variable object.
- Executing a sequence of statements "S1; S2" and a "while" loop are as before.

The execution rules are given in the following pictures:

Implementation

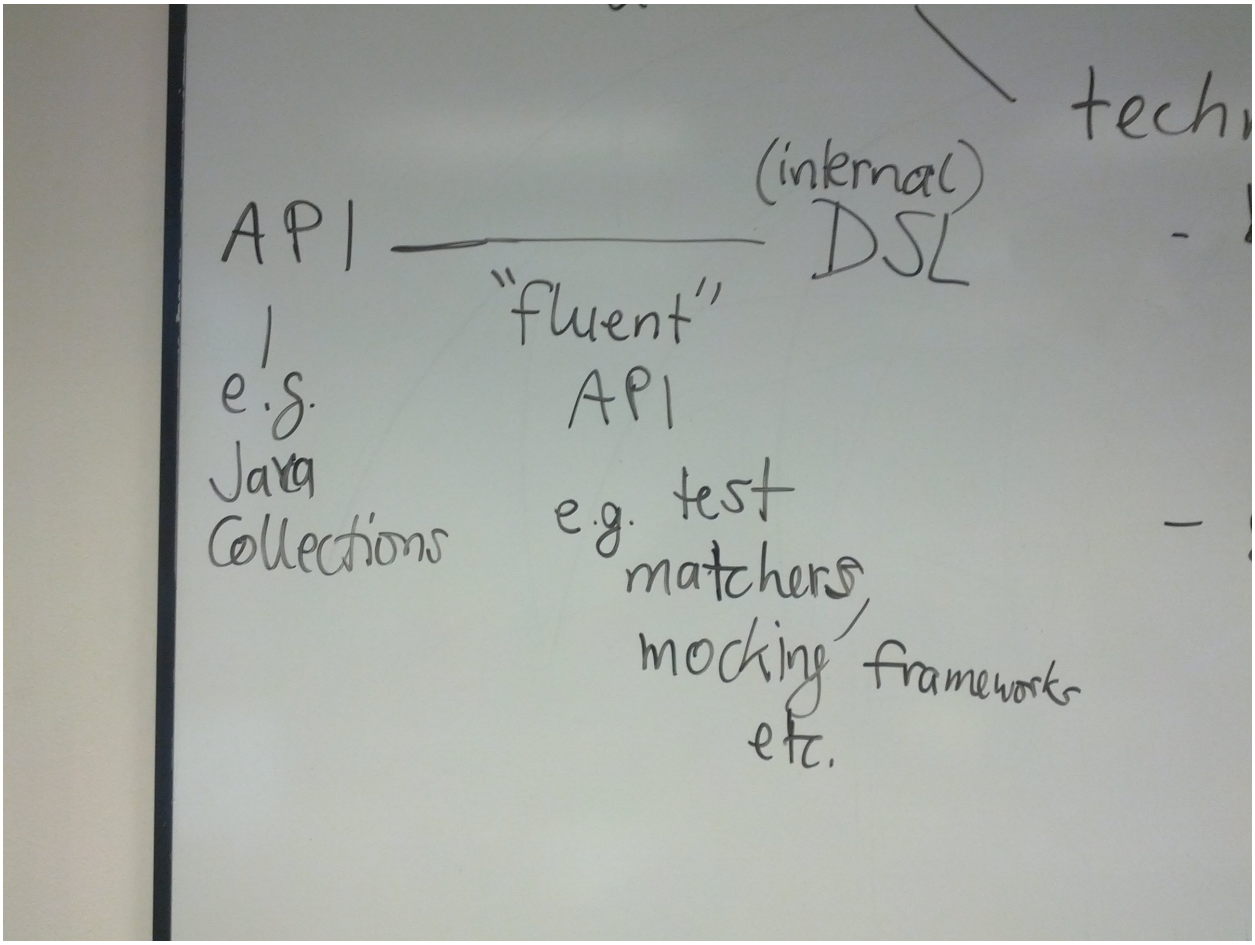
The entire code for the implementation of the simple imperative language with records is available [here](https://github.com/lucproglangcourse/misc-scala/tree/master/src/main/scala/records).

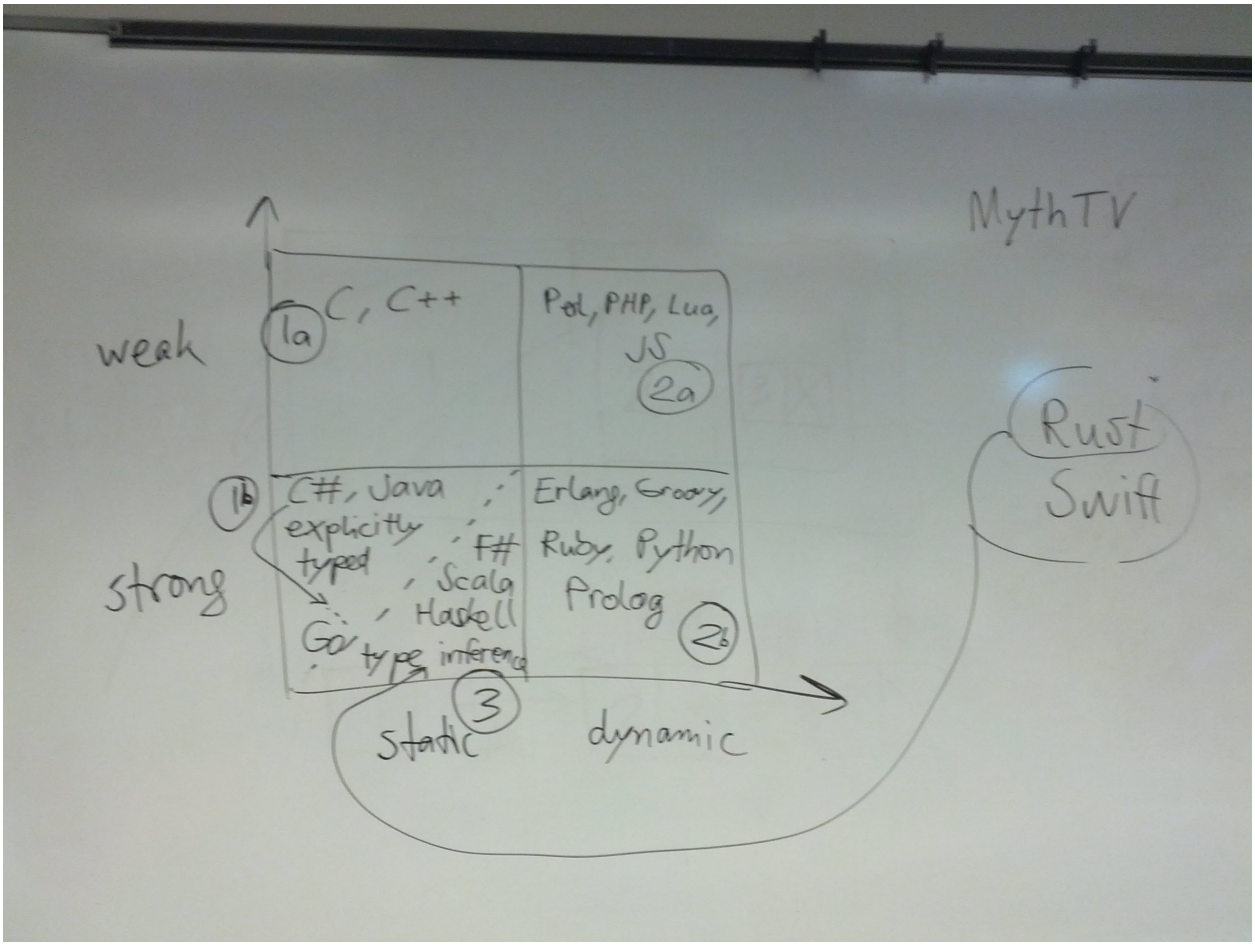
1.4.3 Type systems

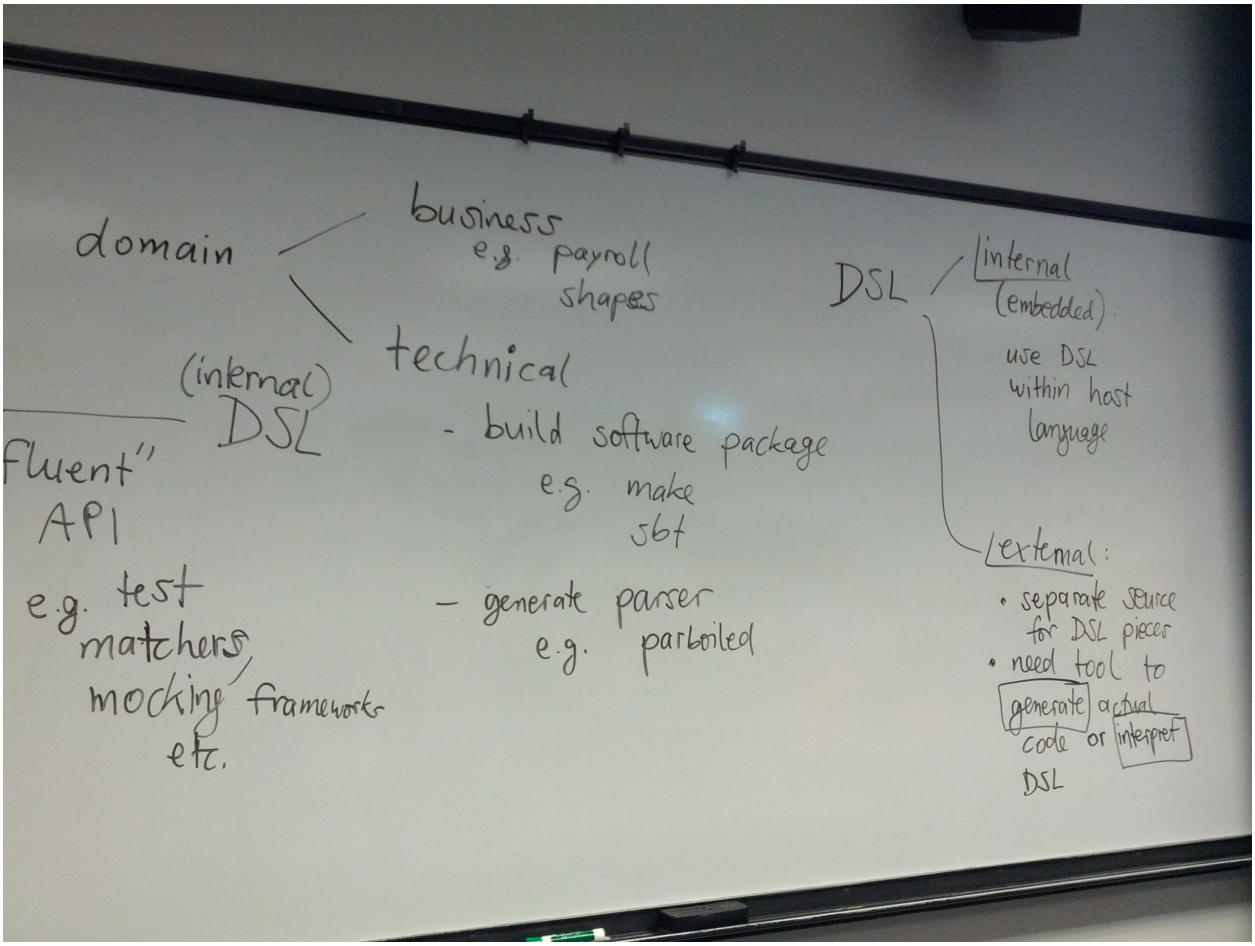
- [design space](http://en.wikipedia.org/wiki/Type_system), tradeoffs, [representative examples](http://en.wikipedia.org/wiki/Comparison_of_type_systems)
- [presentation slides 10-29](http://klaeufer.github.io/luc-amc.html#(10))
- performance implications/[shootout](http://benchmarksgame.alioth.debian.org/u64q/which-programs-are-fastest.php)

1.4.4 Domain-specific languages

- http://c2.com/cgi/wiki?DomainSpecificLanguage
- http://martinfowler.com/bliki/FluentInterface.html
- http://martinfowler.com/bliki/DomainSpecificLanguage.html
- http://martinfowler.com/books/dsl.html
- http://en.wikipedia.org/wiki/Domain-specific_language
- [SE Radio episode 182](http://feedproxy.google.com/~r/se-radio/~3/2VCOnKZ97MU/)




MythTV
weak
strong
1a
C, C++
Pol, PHP, Lua, JS
2a
1b
C#, Java
explicitly typed
F#
Scala
Haskell
Go
type inference
3
Erlang, Groovy, Ruby, Python
Prolog
2b
Rust
Swift
static
dynamic



1.5 Introduction to Concurrency

Todo

add relevant content from CDER chapter

- concurrency overview and foundations - why and when concurrency?
  - when it is a natural fit for the problem domain - multiple autonomous behaviors/simulations - user interfaces: timed events, background activities
  - when the technical solution domain requires it - more efficient use of available resources: asynchronous computing - graphical user interfaces: queuing of low-level input events - multi-core systems - network services/distributed systems
  - fundamentals of concurrency - physical (parallelism) versus logical concurrency - speedup and when to expect it

Lecture

fundamentals of concurrency

- physical (parallelism) versus logical concurrency

- speedup and when to expect it
- data parallelism versus task parallelism

activity terminology and concerns

- process: own memory
- thread: shared memory
- foreground versus background
- CPU-bound [example](https://gist.github.com/klaeufer/5409687) versus IO-bound [example](https://gist.github.com/klaeufer/5409800)
- run-to-completion versus coordination
- progress reporting
- cancelation

projects

- [project 4](https://trello.com/c/KTUBorWF/78-project-4)
- project 3c clinic

Presentation

According to this https://trello.com/c/a7vQg3qp/54-presentation-schedule.

thread safety

- nondeterminism [example](https://gist.github.com/klaeufer/5409800) - extent of nondeterminism: see [CDER chapter](https://docs.google.com/document/d/1QbPEApea_spbemqluf7xPyaTEf_b2ZQYdBNMeiEOHuw)
- race conditions [example](https://gist.github.com/klaeufer/5409867)
- root cause of thread safety problems

dealing with shared state

- mutual exclusion/locking
- confinement
- immutability
- case study: GUIs and the single-threaded rule

(conflicting) design forces

- correctness/(thread-)safety
- liveness/deadlock - dining philosophers [example](http://www.doc.ic.ac.uk/~jnm/book/book_applets/Diners.html) <- *run on Firefox or Safari with site added to Java security settings*
- fairness/starvation
- performance - throughput - latency - jitter

specific concurrency mechanisms

language constructs, patterns, building blocks

- threads (familiar from 313/413)
- monitors: synchronized/locks, wait/notify
- fully synchronized object (pattern/building blocks)

- Android - [AsyncTask](http://developer.android.com/reference/android/os/AsyncTask.html) - [ThreadPoolExecutor](http://developer.android.com/reference/java/util/concurrent/ThreadPoolExecutor.html)
- [java.util.concurrent](https://docs.oracle.com/javase/8/docs/api/java/util/concurrent/package-summary.html) - atomic variables - thread-safe collections - FIFO locks - ...
- [Scala parallel collections](http://docs.scala-lang.org/overviews/parallel-collections/overview.html)
- [futures and promises intro](http://docs.scala-lang.org/overviews/core/futures.html)
- [composable futures in Scala/Akka](http://doc.akka.io/docs/akka/current/scala/futures.html)
- [example: concurrent web requests](https://gist.github.com/klaeufer/3d6a15837bae8d7d5dd07ad9f0db9b97)
- [actors](https://en.wikipedia.org/wiki/Actor_model)
- [reactive streams](http://www.reactive-streams.org/) - see also [Akka streams](http://doc.akka.io/docs/akka/2.4.4/scala/stream)
- [software transactional memory](https://en.wikipedia.org/wiki/Software_transactional_memory)

projects

- [project 4](https://trello.com/c/KTUBorWF/78-project-4)
- project 3c and 4 clinic

Presentation

According to this https://trello.com/c/a7vQg3qp/54-presentation-schedule.

References

Concurrent and asynchronous computing

- Läufer and Thiruvathukal, [CDER chapter](https://docs.google.com/document/d/1QbPEApea_spbemqluf7xPyaTEf_b2ZQYdBNMeiEOHuw) (especially the section on fundamentals of thread safety)
- Goetz et al., [JCIP](http://jcip.net.s3-website-us-east-1.amazonaws.com/)
- Doug Lea, [CPJ](http://gee.cs.oswego.edu/dl/cpj)
- Thiruvathukal and Christopher, [HPJPC](https://code.google.com/p/hpjpc/)
- [SE Radio episode on concurrency: part 1](http://feedproxy.google.com/~r/se-radio/~3/lJLihLsyf0M/)
- [SE Radio episode on concurrency: part 2](http://feedproxy.google.com/~r/se-radio/~3/Wh7E6YT1_JI/)
- [SE Radio episode on concurrency: part 3](http://feedproxy.google.com/~r/se-radio/~3/WZ7h3kzRARY/)
- [SE Radio episode on concurrency: part 4](http://feedproxy.google.com/~r/se-radio/~3/bqICWnvwuGw/)
- [futures and promises overview](http://arild.github.io/scala-workshop/#/12)
- [RxJava/RxScala](https://github.com/ReactiveX/RxScala)
- [asynchronous programming video](https://www.youtube.com/watch?v=UKjcJ13bD3s)
- [reactive/asynchronous programming with RxJava/RxScala video](https://www.youtube.com/watch?v=_t06LRX0DV0)

1.6 Summary

summary/recap goes here

CHAPTER

TWO

APPENDIX

2.1 Programming Language Design Principles

This is the list of programming language design principles from Bruce MacLennan's excellent Principles of Programming Languages (2nd edition, 1986, ISBN 0-03-005163-0, found on Oliver Wyman's blog.

- **Abstraction**: Avoid requiring something to be stated more than once; factor out the recurring pattern.
- **Automation**: Automate mechanical, tedious, or error-prone activities.
- **Defense in Depth**: Have a series of defences so that if an error isn't caught by one, it will probably be caught by another.
- **Information Hiding**: The language should permit modules designed so that (1) the user has all of the information needed to use the module correctly, and nothing more; and (2) the implementor has all of the information needed to implement the module correctly, and nothing more.
- **Labeling**: Avoid arbitrary sequences more than a few items long. Do not require the user to know the absolute position of an item in a list. Instead, associate a meaningful label with each item and allow the items to occur in any order.
- **Localized Cost**: Users should only pay for what they use; avoid distributed costs.
- **Manifest Interface**: All interfaces should be apparent (manifest) in the syntax.
- **Orthogonality**: Independent functions should be controlled by independent mechanisms.
- **Portability**: Avoid features or facilities that are dependent on a particular machine or a small class of machines.
- **Preservation of Information**: The language should allow the representation of information that the user might know and that the compiler might need.
- **Regularity**: Regular rules, without exceptions, are easier to learn, use, describe, and implement.
- **Security**: No program that violates the definition of the language, or its own intended structure, should escape detection.
- **Simplicity**: A language should be as simple as possible. There should be a minimum number of concepts, with simple rules for their combination.
- **Structure**: The static structure of the program should correspond in a simple way to the dynamic structure of the corresponding computations.
- **Syntactic Consistency**: Similar things should look similar; different things different.
- **Zero-One-Infinity**: The only reasonable numbers are zero, one, and infinity.

2.2 Resources

Todo

consider moving most of these into the corresponding chapters as "further reading"

2.2.1 Key resources

- Code examples on GitHub
- Instructor's videos on YouTube
- External playlist on YouTube
- External references/bookmarks on Diigo
- General resources
- Coding guidelines
- Library of research resources (LRR)
- Loyola library's CS subject guide
- Some professional meetups around town

2.2.2 Code examples

The code examples for this course are at https://github.com/lucproglangcourse. (The non-Scala ones are much less likely to change.)

The recommended process for exploring an example is to fork it into your own GitHub account. Please create issues and/or submit pull requests with improvements or corrections.

It is important for you to receive notifications of changes to the key examples.

- These will automatically appear on this Slack channel, which you can join.
- If you have a GitHub account, you can instead receive notifications within GitHub by watching and/or starring the repositories of interest directly.

2.2.3 Required text

- **Programming in Scala, Third Edition** *By: Martin Odersky; Lex Spoon; Bill Venners* Publisher: Artima Press Pub. Date: April 28, 2016 ISBN-10: 0-9815316-8-7 ISBN-13: 978-0-9815316-8-7 available free on Safari

2.2.4 Recommended/supplementary texts

- **Scala Test-Driven Development** *By: Gaurav Sood* Publisher: Packt Publishing Pub. Date: October 27, 2016 Web ISBN-13: 978-1-78646-302-9 Print ISBN-13: 978-1-78646-467-5 available free on Safari
- **Basics of Compiler Design** *By: Torben Mogensen* available online
- **Learning Concurrent Programming in Scala** *By: Aleksandar Prokopec* Publisher: Packt Publishing Pub. Date: November 28, 2014 Print ISBN-13: 978-1-78328-141-1 Web ISBN-13: 978-1-78328-142-8 available free on Safari

- **Seven Concurrency Models in Seven Weeks** *By: Paul Butcher* Publisher: Pragmatic Bookshelf Pub. Date: June 30, 2014 Print ISBN-13: 978-1-937785-65-9 available free on Safari

2.2.5 Additional texts on Scala

- **Learning Scala** *By: Jason Swartz* Publisher: O'Reilly Media, Inc. Pub. Date: December 24, 2014 Print ISBN-13: 978-1-4493-6793-0 available free on Safari
- **A Concise Introduction to Scala** *By: David Matuszek* available online
- **Scala for the Impatient** *By: Cay S. Horstmann* Publisher: Addison-Wesley Professional Pub. Date: March 06, 2012 Print ISBN-10: 0-321-77409-4 Print ISBN-13: 978-0-321-77409-5 Web ISBN-10: 0-13-276177-7 Web ISBN-13: 978-0-13-276177-2 first half available free from Typesafe
- **Scala for Java Developers** *By: Thomas Alexandre* Publisher: Packt Publishing Pub. Date: April 25, 2014 Print ISBN-13: 978-1-78328-363-7 Web ISBN-13: 978-1-78328-364-4 available free on Safari
- **Functional Programming in Scala** (advanced) *By: Paul Chiusano and Rúnar Bjarnason* Foreword by Martin Odersky Publisher: Manning Publications Pub. Date: August 2014 Print ISBN: 9781617290657 available free on Safari

2.2.6 Online Scala resources

- SIGCSE Scala workshop
- The Neophyte's Guide to Scala
- Twitter Scala School
- Scala resources from an Applied Text Analysis course
- Scala documentation site
- Scala API documentation
- Scala style guide
- Scala parser combinator tutorial
- parboiled2 parser generator
- scalaz API documentation (advanced)
- scalaz cheat sheet (advanced)

2.2.7 Resources on program representation and interpretation

- Compiler design tutorial
- Abstract syntax tree
- Regular expressions
- Context-free grammars (CFGs)
- Extended BNF
- Parser generators
- Parser combinators
- Parsing expression grammars (PEGs)

- Interpreters

2.2.8 Other resources

- Bruce Maclennan's programming language design principles
- Seven languages in seven weeks
- 10 programming languages that could shake up IT
- 7 programming languages on the rise
- *Growing a Language* by Guy Steele: video PDF
- *Origami Programming* by Jeremy Gibbons: PDF
- *On Understanding Types, Data Abstraction, and Polymorphism* by Cardelli and Wegner PDF
- Learn You a Haskell for Great Good!
- **Masterminds of Programming, 1st Edition** *By: Federico Biancuzzi; Shane Warden* Publisher: O'Reilly Media, Inc. Pub. Date: March 27, 2009 Print ISBN-13: 978-0-596-51517-1 available free on EBL
- **Programming Language Pragmatics** (3rd ed.) *By: Michael L. Scott* ISBN: 978-0-12-374514-9 Pub date: Mar 23, 2009 Elsevier Science & Technology companion site
- **Functional Thinking** *By: Neal Ford* Publisher: O'Reilly Media, Inc. Pub. Date: July 10, 2014 Print ISBN-13: 978-1-4493-6551-6 available free on Safari
- **Real World Haskell** *By: Bryan O'Sullivan, Don Stewart, and John Goerzen* Publisher: O'Reilly Media. Pub. Date: November 2008 Print ISBN-10: 0596514980 Print ISBN-13: 978-0596514983 available online for free
- **Haskell: The Craft of Functional Programming (3rd Edition)** *By: Simon Thompson* Publisher Addison-Wesley Professional Pub. Date: 2012 Print ISBN-13: 9780201882957 online companion site

---

Todo

master list of languages for presentation

2.3 Course Software

You can choose either or both of the following options.

2.3.1 Cloud-based development environment

This option is recommended if you have any problems with your local setup or if you prefer a basic cloud-based setup.

Pros

- consistent, cloud-based environment

Cons

- requires good network connection
- lacks code completion

Setup

- Create a GitHub account if you don't have one already.
- Use your GitHub account to log into Codenvy.
- Create a new workspace with the following settings:
  - New from blank ...
  - Create new workspace from stack > BLANK
  - Name: lucproglangcourse
  - RAM: 3GB
- Wait for the workspace to become available. Now you have a persistent, cloud-based, standard Ubuntu/Linux command-line environment that you can work in and come back to from any computer.
- In the resulting Linux terminal, run this gist to install the Scala build tool (sbt):

```
wget https://gist.github.com/klaeufer/c4040de09f51395b96be/raw/install-sbt-ubuntu.
↪sh && sh install-sbt-ubuntu.sh
```

- Now you can check out existing projects using git or create a new one like so:

```
$ mkdir -p hello/src/main/scala
$ cd hello
$ cat > src/main/scala/Hello.scala
object Hello extends App {
  println("hello")
}
^D
$ sbt run
...
hello
```

2.3.2 Locally installed development environment

This option will give you an advanced development environment with code completion, type info, etc.

Pros

- provides powerful capabilities, including code completion
- does not require network connection

Cons

- might be slow on older machines
- need to maintain on each machine you use

Required components

- Java 8 and/or newer JDK
- Git DSCM (usually preinstalled on Mac OS and Linux)
  - recommended installation option on Windows: *Use Git and optional Unix tools from the Windows Command Prompt*
- sbt
- IntelliJ IDEA CE
  - check specific prerequisite details for your platform
  - for the following steps, make sure you have no projects open and are looking at the welcome window as in the attached screenshot
  - *Scala plugin installation:* IntelliJ IDEA > Configure (bottom right) > Plugins > Browse repositories > find and right-click Scala > download and install > close repository browser > OK to restart IDEA
  - *JDK configuration*: IntelliJ IDEA > Configure > Project Defaults > Project Structure > Platform Settings > SDKs > + > JDK > navigate to the installation directory of your most recent JDK > OK

Optional for Windows users

- TortoiseGit (integration of Git with Windows Explorer)
- SourceTree Git client

Optional for Mac users

- SourceTree Git client

Optional for all IntelliJ users

- These are useful Android Studio/Intellij IDEA plugins. (Installation procedure is the same as for the Scala plugin.)
  - Key Promoter (helps you learn keyboard shortcuts)
  - MultiMarkdown
- IntelliJ IDEA Scala worksheets are a convenient way to do REPL-style exploratory programming while being able to modify and save all your code.

  Check out this example, especially the simple option and list examples.

  Worksheets can live in any folder and coexist with other code. So you can start exploring something in a worksheet and then move it into your production code.

2.3.3 sbt tips and optional plugins

This section applies to all uses of sbt, whether cloud-based or on the local command line.

In conjunction with a text editor, sbt's triggered execution for testing will significantly shorten the edit-compile-run/test cycle.

These are useful additional sbt plugins. You can install them per project or globally. Some of the example projects already come with one or more of these plugins.

- sbt-scoverage: uses Scoverage to produce a test code coverage report
- sbt-updates: checks central repos for dependency updates
- ls-sbt: browse available libraries on GitHub using ls.implicit.ly
- sbt-dependency-graph: creates a visual representation of library dependency tree
- cpd4sbt: copy/paste detection for Scala
- scalastyle: static code checker for Scala
- sbt-stats: simple, extensible source code statistics/metrics
- sbt-scalariform: automatic source code formatting using Scalariform

2.3.4 Required for remote participation

- Adobe Connect for holding class remotely in case of weather emergencies

2.4 Course Syllabus

This is the official course syllabus for the current section(s) of this course.

2.4.1 Course: COMP 372/471 Programming Languages

- Prerequisite: Comp 313
- Official course description: Comp 372 | Comp 471

2.4.2 Section: 001 Spring 2017

- *Instructor:* Konstantin Läufer
- *Class time and location (Spring 2017)*: Tue 14:45-17:15, Mundelein Center 607, LSC
- *Office hour:* Tue 17:15-18:00 in Mundelein Center 607, LSC, and by appointment
- *Required text:* **Programming in Scala, Third Edition** *By: Martin Odersky; Lex Spoon; Bill Venners* Publisher: Artima Press Pub. Date: April 28, 2016 ISBN-10: 0-9815316-8-7 ISBN-13: 978-0-9815316-8-7 available free on Safari
- Grading (tentative): - 45% quizzes & exams - 50% projects & presentations *(Percentage effort on each group project will be measured by an end-of-term questionnaire. Group project grades and/or final course grades may be adjusted to account for significant discrepancies in effort among group members.)* - 5% participation (in-class and online, including announcements of and reports from relevant professional events, GitHub issues and PRs for course examples, etc.)

- Ground rules
- Sakai site for this section (gradebook)
- Slack team *(mandatory subscription)*

2.4.3 Detailed Course Outline

- functional programming (6 weeks)
  - defining algebraic data types
    - scalars: enumerations
    - sublinear structures: numbers, option
    - linear structures: lists, maps
    - nonlinear structures: trees
  - implementing behaviors on algebraic data types
    - pattern matching
    - recursion
  - higher-order functions
  - predefined types and their behaviors
  - recursion patterns
  - higher-kinded types
- programming language representation and interpretation/execution (5 weeks)
  - language design space
  - scanners (lexical analyzers)
  - parsers (syntax analyzers)
  - interpreters and compilers
  - domain-specific languages
- concurrent and parallel programming (4 weeks)
  - parallel collections
  - futures and promises
  - progress reporting and cancelation
  - asynchronous programming/reactive extensions (Rx)
  - advanced mechanisms
    - explicit threads
    - actors
    - software-transactional memory
    - task-parallel library
    - functional data structures

2.4.4 Key Resources

- Code examples on GitHub
- Instructor's videos on YouTube
- External playlist on YouTube
- External references/bookmarks on Diigo
- General resources
- Coding guidelines
- Library of research resources (LRR)
- Loyola library's CS subject guide
- Some professional meetups around town

Todo

add important dates

Todo

move ground rules here

2.5 TODO

Todo

incorporate visual presentation here

(The original entry is located in /Users/laufer/Dropbox/Home/Work/writing/lucproglangcourse/source/overview.rst, line 41.)

Todo

map Bloom levels to course outline

(The original entry is located in /Users/laufer/Dropbox/Home/Work/writing/lucproglangcourse/source/overview.rst, line 61.)

Todo

add relevant content from CDER chapter

(The original entry is located in /Users/laufer/Dropbox/Home/Work/writing/lucproglangcourse/source/concurrency.rst, line 5.)

Todo

consider moving most of these into the corresponding chapters as "further reading"

(The original entry is located in /Users/laufer/Dropbox/Home/Work/writing/lucproglangcourse/source/resources.rst, line 5.)

Todo

master list of languages for presentation

(The original entry is located in /Users/laufer/Dropbox/Home/Work/writing/lucproglangcourse/source/resources.rst, line 193.)

Todo

add important dates

(The original entry is located in /Users/laufer/Dropbox/Home/Work/writing/lucproglangcourse/source/syllabus.rst, line 48.)

Todo

move ground rules here

(The original entry is located in /Users/laufer/Dropbox/Home/Work/writing/lucproglangcourse/source/syllabus.rst, line 50.)

Todo

add CI/CD using Travis

(The original entry is located in /Users/laufer/Dropbox/Home/Work/writing/lucproglangcourse/source/index.rst, line 48.)

Todo

figure out why search isn't working

(The original entry is located in /Users/laufer/Dropbox/Home/Work/writing/lucproglangcourse/source/index.rst, line 50.)

Todo

add sample projects, activities, tests, and (tiered) master list for presentations

(The original entry is located in /Users/laufer/Dropbox/Home/Work/writing/lucproglangcourse/source/index.rst, line 52.)

Todo

add stuff from https://trello.com/c/kAMGZ4uf/28-add-in-next-iteration

(The original entry is located in /Users/laufer/Dropbox/Home/Work/writing/lucproglangcourse/source/index.rst, line 54.)

Todo

add CI/CD using Travis

Todo

figure out why search isn't working

Todo

add sample projects, activities, tests, and (tiered) master list for presentations

Todo

add stuff from https://trello.com/c/kAMGZ4uf/28-add-in-next-iteration

CHAPTER

THREE

INDICES AND TABLES

- genindex
- modindex
- search